

Question				Marking details	Marks Available															
					AO1	AO2	AO3	Total	Maths	Prac										
5	(a)	(i)	I	$X^D Y$ $X^D X^d$ $X^D Y$ $X^D X^d$ All four correct = 2 marks 2/3 correct = 1 mark 0/1 correct = 0 marks		2		2												
			II	<table><tr><th>Offspring genotypes</th><th>Offspring phenotype</th></tr><tr><td>$X^D X^D$</td><td>Normal female</td></tr><tr><td>$X^D X^d$</td><td>Normal female</td></tr><tr><td>$X^D Y$</td><td>Normal male</td></tr><tr><td>$X^d Y$</td><td>X-SCID male</td></tr></table> All correct = 2 marks 2/3 correct = 1 mark 0/1 correct = 0 marks Accept 'unaffected'	Offspring genotypes	Offspring phenotype	$X^D X^D$	Normal female	$X^D X^d$	Normal female	$X^D Y$	Normal male	$X^d Y$	X-SCID male		2		2		
Offspring genotypes	Offspring phenotype																			
$X^D X^D$	Normal female																			
$X^D X^d$	Normal female																			
$X^D Y$	Normal male																			
$X^d Y$	X-SCID male																			
		(ii)		64 cases = 2 marks If incorrect award 1 mark for <u>3139000</u> 48933 60 / 60.0 / 64.1 / 64.15 (incorrect sf)		2		2	2											